Measurement Invariance of the DDI Dermatology Benchmark Across Fitzpatrick Skin Type

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Abstract

Fairness audits of dermatology AI typically compare accuracy across Fitzpatrick Skin Type (FST) groups—a six-level dermatologic scale of constitutive skin pigmentation, conventionally collapsed to FST-I/II (lightest), III/IV, and V/VI (darkest)—and treat the gap as direct evidence of differential model capability. This inference confuses three distinct phenomena: differences in capability, differences in item difficulty, and differences in how items function as measurements across groups. We propose to disentangle these by combining amortized Item Response Theory (IRT) calibration of foundation vision-language models against the Diverse Dermatology Images (DDI) benchmark with item-level measurement-invariance analysis grouped by FST. Our primary endpoint is an aggregate FST difficulty shift on the Rasch logit scale, with a pre-specified *falsification threshold*—a magnitude floor below which we will declare no meaningful non-invariance even if a statistical test rejects—of $|\Delta b_{V\text{-}VI-I\text{-}II}| \geq 0.5$ logits, conjoined with a bootstrap 95% CI excluding zero. A Linear Logistic Test Model (LLTM) decomposes whether residual FST signal survives controls for lesion category and malignancy.

1 Introduction

Dermatology AI tools are increasingly deployed in clinical care, with a wider range of applications under active consideration [Daneshjou et al., 2022]. Fitzpatrick-Skin-Type-stratified accuracy has become the dominant fairness metric in those evaluations [Daneshjou et al., 2022, Groh et al., 2021], and regulators, procurement reviewers, and journals now treat subgroup gaps as direct evidence of inequitable model capability. If the metric is validity-blind, the consequence is two-sided: tools whose subgroup gap reflects benchmark composition rather than capability are penalized unfairly, and tools whose lack of a stratified gap conceals true capability differences are cleared unjustifiably. Both errors are borne disproportionately by the patients the audits are intended to protect, which makes the question of whether stratified accuracy actually measures what it claims to measure a deployment-grade concern, not just a measurement-theory one.

Suppose an FST-V/VI subgroup exhibits lower accuracy than an FST-I/II subgroup. Three explanations are observationally indistinguishable from a stratified metric. *Capability difference*: models are genuinely worse at the underlying construct on darker skin. *Difficulty composition*: the FST-V/VI items happen to be harder along construct-relevant dimensions (rarer lesions, more advanced disease), independent of FST per se. Critically, DDI is not paired across FST: the lesion-category distribution differs across strata, so a per-stratum mean accuracy compares partially non-overlapping mixtures of lesions, and a stratified gap admits a difficulty-composition explanation that cannot be ruled out without item-level analysis. *Measurement non-invariance*: items in the two strata do not measure the same construct in the same way, so the accuracy numbers are not on a common scale [Borsboom et al., 2004]. Without separating these, any equity claim derived from a subgroup gap is unsupported.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

Do items in the DDI benchmark [Daneshjou et al., 2022] exhibit measurement non-invariance across FST groups when foundation vision-language models are evaluated on them, and if so, is that non-invariance attributable to clinical factors (lesion category, malignancy) rather than FST per se?

We adapt amortized IRT calibration [Truong et al., 2025] to image-classification benchmarks, combine it with measurement-invariance analysis to produce per-item validity diagnostics, and apply the pipeline to DDI. We are deliberately narrow. DDI is sourced entirely from a single institution (Stanford), so our findings speak to measurement properties of this benchmark rather than to dermatology AI evaluation in general, and any institution-level acquisition effects are absorbed into the FST contrast rather than partialed out. We do not causally identify FST as an independent driver of difficulty, since FST and lesion category are structurally entangled in DDI and further confounded with acquisition site. We report partial-identification analyses and adopt a hedged consequential-validity framing.

2 Related work

IRT for ML benchmark evaluation. Truong et al. [2025] introduced amortized IRT calibration for text benchmarks; we adopt their approach, port it to image embeddings, and extend it to measurement invariance, which they do not address. Polo et al. [2024] and Rodriguez et al. [2021] use IRT to construct compact benchmark subsets and rank NLP leaderboards, respectively; neither analyzes subgroup invariance.

DIF and measurement invariance. *Differential item functioning* (DIF) is the psychometric term for a test item whose probability of a correct response, conditional on the latent trait being measured, varies across groups of respondents—i.e., an item that does not function as the same measurement instrument across groups. Holland and Wainer [1993] provides the canonical DIF reference (Mantel–Haenszel, standardization), and Millsap [2011] grounds the configural/metric/scalar-invariance hierarchy we adopt. We invert the usual direction: rather than treating DIF as a property of human test items, we use it as a tool to audit ML benchmarks.

Fairness and validity in medical AI. Daneshjou et al. [2022] introduced DDI to enable FST-stratified evaluation and documented substantial accuracy gaps; subsequent work has expanded the catalogue [Groh et al., 2021]. Wang et al. [2025] argue more broadly that fairness metrics conflate disparate kinds of difference. We are not aware of prior work applying item-level measurement-invariance analysis to medical AI benchmarks; we close that gap.

3 Methods

Course-material connections. The pipeline below draws on three threads from Truong and Koyejo [2026]. The Rasch model and its amortized calibration variant are the item-response and amortized-factor models of Chapter 2 (Foundations of Measurement). Joint MLE for θ and ϕ follows the parameter-estimation treatment of Chapter 3 (Learning). The aggregate measurement-invariance test (§3) operationalizes Chapter 6's framing of differential item functioning and construct-irrelevant variance as validity diagnostics, with the construct/criterion validity vocabulary used as the consequential frame. The LLTM decomposition (§3) is a structured-latent-variable extension that bridges Chapters 2 and 6, casting the residual-FST coefficient as a bridge between benchmark instrumentation and validity audit.

Respondent selection. We plan to query 12–20 foundation vision and vision-language models spanning architectural families. Contrastive: CLIP, SigLIP, BiomedCLIP. Self-supervised: DINOv3. Open multimodal: LLaVA-class, Qwen-VL, InternVL. Closed APIs: GPT-4o-class, Claude, Gemini. Classical IRT assumes exchangeable respondents; foundation models share pretraining corpora and architectures and are not random draws from any population. We therefore weaken the population interpretation, and ability estimates should be read as referring to the currently extant frontier multimodal ecosystem. The primary endpoint (below) is an aggregate item-level statistic robust to this weakening.

Query protocol. Per (model, item): single response, binary task framing (benign vs. malignant), single fixed prompt version-controlled in `config/protocol.yaml`, temperature 0 where supported, structured-output extraction with a GPT-4o-mini fallback parser. *Refusals are an analytic outcome category, not coerced into a default label.* A paraphrase-perturbed subset measures prompt sensitivity; $K = 3$ samples per (model, item) on a stratified subset estimate API non-determinism.

Amortized Rasch calibration. Let $X_{ij} \in \{0, 1\}$ indicate whether model j answered item i correctly. Under the Rasch model, $P(X_{ij} = 1 \mid \theta_j, b_i) = \sigma(\theta_j - b_i)$ [Truong and Koyejo, 2026, Ch. 2]. Following Truong et al. [2025], we amortize item difficulty as $b_i = f_\phi(e_i)$ where e_i is a fixed image embedding and f_ϕ is a small MLP. Primary embedding model: **BiomedCLIP** [Zhang et al., 2025] (dermatology-relevant pretraining). **DINOv3** [Siméoni et al., 2025] is used in parallel as a domain-agnostic robustness check. We jointly optimize $\{\theta_j\}$ and ϕ by maximizing the masked binary log-likelihood with Adam. Validation: (i) held-out item predictive AUC (20% items); (ii) Spearman correlation against traditional joint-MLE Rasch on items with sufficient response density; (iii) infit/outfit mean-squares with flagging threshold $[0.7, 1.3]$.

Aggregate measurement invariance (primary endpoint). Standard DIF treats persons as the grouping variable [Truong and Koyejo, 2026, Ch. 6]; in our setup FST is an item-level attribute, inverting that configuration. We therefore frame the primary analysis as a measurement-invariance test on item subsets. After fitting the amortized Rasch model, we residualize $\hat{b}_i$ against lesion category and malignancy by OLS and compute

$$\Delta = \overline{\hat{b}_i^{\text{resid}}}_{i \in \text{FST V–VI}} - \overline{\hat{b}_i^{\text{resid}}}_{i \in \text{FST I–II}}. \tag{1}$$

Uncertainty: non-parametric bootstrap over items, $B = 2000$ resamples. **Decision rule**:

$$|\Delta| \geq 0.5 \text{ logits} \quad \text{AND} \quad \text{bootstrap 95\% CI of } \Delta \text{ excludes zero} \quad \Rightarrow \quad \text{meaningful FST non-invariance.} \tag{2}$$

The 0.5-logit threshold reflects the conventional small-DIF cutoff in psychometric practice; we additionally report 0.3 and 0.7 as sensitivity checks but treat 0.5 as binding. *Configural-invariance check*: we refit the amortized model separately on FST-I/II and FST-V/VI subsets and report the Spearman correlation of model abilities; $\rho > 0.9$ is consistent with configural invariance (same construct ranks models the same way) even when difficulties shift.

LLTM mechanism decomposition and metadata-only baselines. We fit nested specifications of the form $b_i = \beta_0 + \boldsymbol{\beta}^\top \mathbf{x}_i$ with increasing feature sets: **M1** lesion category; **M2** +malignancy; **M3** +FST group dummies (I/II reference). M1 and M2 double as *metadata-only baselines*: they predict per-item difficulty from clinical attributes that any reader of the dataset card could enumerate without running a single model query, and their R^2 against $\hat{b}_i$ is the share of difficulty variance that a chart-review-style audit would already explain. The headline mechanism estimate is $\boldsymbol{\beta}_{\text{FST}}$ in M3: the residual FST coefficient after lesion and malignancy controls. We report (i) R^2 at each nesting step so the marginal contribution of the amortized embedding signal above metadata is explicit (the gap between R^2_{M2} and an embedding-only regression R^2_{Emb} is the finer-grained-insight contribution of IRT calibration), (ii) bootstrap 95% CIs on all coefficients, and (iii) the lesion-category $\times$ FST contingency table that drives the entanglement (§4). We do not include image-acquisition covariates in the LLTM because the DDI release does not expose acquisition metadata (device, site, photographer, capture protocol); residual acquisition confounding is treated qualitatively in the limitations.

Mechanism and refusal-rate DIF. (i) Spearman correlations of per-item residualized difficulty with image-derived properties computable from the JPEG (resolution, mean luminance, color-channel statistics) and with embedding-space distance to the FST-I/II centroid; these are descriptive, not a quantitative decomposition. (ii) Refusal probability by FST via logistic regression with cluster-bootstrap CI clustered on model. A refusal pattern that loads on FST is itself evidence of differential item functioning, and we report it as such.

Respondent-side analysis: family-stratified ability and contrast (exploratory). The respondent non-exchangeability noted in §3 is itself analyzable. We annotate each model in `config/models.yaml` with an architectural family (contrastive image-text, self-supervised image, open multimodal-instruct, closed multimodal-instruct) and treat family as a grouping variable on the respondent side. We report: (a) the within-family mean and standard deviation of $\hat{\theta}_j$ and a one-way ANOVA F-statistic for family on $\hat{\theta}$, partitioning between-family from within-family variance in ability; (b) the family-stratified primary contrast Δ_{family}, i.e., the FST difficulty shift recomputed with the respondent set restricted to each family in turn, so a finding that the V–VI vs. I–II gap is concentrated in a particular family (e.g., contrastive image-text models) is read off directly; (c) the Spearman correlation between per-item difficulty estimates obtained from disjoint respondent subsets defined by family, which tests whether different backbones produce a common item-difficulty ranking or partition into incompatible measurement instruments. This block is explicitly exploratory: with $J = 12$–20 models distributed across four families, within-family sample sizes are small and any family-level Δ carries wide uncertainty, so we report bootstrap CIs and decline a falsification threshold here.

4 Experimental setup

Data: DDI [Daneshjou et al., 2022] as the primary benchmark with FST I/II, III/IV, V/VI labels. We will report the lesion-category $\times$ FST and malignancy $\times$ FST contingency tables on the full DDI release up front, so the reader can read the entanglement structure (§3) off the data before seeing any IRT output. We do not use ISIC for inferential analysis (site-level photographic protocol differences would confound any FST signal); if used, ISIC supplies only pretraining signal for the amortized difficulty MLP in a two-stage design declared before the DDI fit. **Compute:** embedding extraction and open-weights inference run locally on a single GPU; closed-API inference scales as $J_{\text{api}} \times 656 \times P$ for P protocol variants. **Bootstraps:** $B = 2{,}000$ resamples for the primary CI; cluster-bootstrap on models for the refusal logit. **Metrics reported:** held-out Rasch AUC and infit/outfit; amortized-vs-traditional Spearman; primary Δ with bootstrap CI and configural-invariance Spearman; LLTM variance-explained at each nesting level (with M1/M2 read as metadata-only baselines) and $\boldsymbol{\beta}_{\text{FST}}$ CI in M3; mechanism Spearmans; refusal-rate logit with cluster-bootstrap CI; family-stratified $\hat{\theta}$ and family-stratified Δ (§3).

5 Preliminary results (closed-API subpanel)

We report preliminary descriptive and IRT-calibration statistics from the five respondents that have completed the full DDI panel at the time of writing: two closed-API generative models (GPT-4o, Claude Sonnet 4.5) and three contrastive zero-shot models (CLIP ViT-L/14, SigLIP-L/16, Biomed-CLIP). The open-weight generative VLM panel is still being collected; the configural-invariance check (§3) and the LLTM decomposition (§3) are deferred to the full panel. We do, however, run the pre-registered amortized Rasch primary endpoint on the current panel as a smoke test of the pipeline against real data, and the result is informative both for the audit substance and for the validity of the IRT framework itself.

Dataset entanglement (DDI metadata only). Before any model responses are introduced, the structural entanglement that motivates the LLTM decomposition (§3) is visible directly in DDI's metadata. The 656 DDI items span 76 distinct lesion-category labels, and the marginal distribution differs sharply across FST strata. Selected examples: basal-cell carcinoma is concentrated in FST-I/II and FST-III/IV (7 and 34 items respectively, vs. *zero* in V–VI); dysplastic nevus is 2 / 14 / 0; mycosis fungoides is 3 / 3 / 26 (heavily concentrated in V–VI); eczema-spongiotic-dermatitis is 0 / 0 / 4 (V–VI only). A per-FST accuracy aggregate therefore compares partially non-overlapping mixtures of lesion categories, not a paired comparison on a common item pool. The full 76$\times$3 contingency table is reported in `artifacts/contingency/lesion_x_fst.tex`.

The malignancy base rate also varies non-monotonically across FST groups (Table 1): 23.6% in FST-I/II, 30.7% in FST-III/IV, and 23.2% in FST-V/VI. This pattern is incompatible with the simplest "FST-V/VI has more advanced disease" interpretation of an accuracy gap, since by-stratum prevalence peaks in III–IV.

Baseline FST-stratified accuracy. Table 2 reports per-model accuracy under the binary benign-vs-malignant protocol, stratified by FST group. The two generative respondents (GPT-4o, Claude) show V–VI minus I–II gaps of -5 to -7 percentage points, with a non-monotonic pattern across FST (III–IV intermediate or slightly higher than I–II rather than a linear decline). BiomedCLIP

Table 1: Malignancy $\times$ FST contingency on the full DDI release. The non-monotonic per-stratum malignancy base rate (peaking in III–IV) rules out a simple prevalence-shift explanation for the V–VI accuracy gap.

Malignant	FST I–II	FST III–IV	FST V–VI	Total
False	159	167	159	485
True	49	74	48	171
Total	208	241	207	656

Table 2: Baseline FST-stratified accuracy (four-respondent preliminary panel: 2 closed-API generative, 2 contrastive zero-shot). Per-model proportion of items answered correctly under the binary benign-vs-malignant protocol. The right-most column is the V–VI minus I–II accuracy gap that conventional fairness audits would report; the pre-registered Rasch-difficulty analysis (§3) tests whether this gap reflects differential capability or differential item difficulty. Item counts per FST group in parentheses.

Model	N	FST I–II	FST III–IV	FST V–VI	Gap (V–VI $-$ I–II)
GPT-4o	656	0.596 (208)	0.631 (241)	0.541 (207)	-0.055
Claude Sonnet 4.5	656	0.611 (208)	0.602 (241)	0.546 (207)	-0.065
CLIP ViT-L/14	656	0.721 (208)	0.647 (241)	0.691 (207)	-0.030
BiomedCLIP	656	0.562 (208)	0.598 (241)	0.517 (207)	-0.046

SigLIP-L/16 omitted from this table pending verification of full coverage (see §5); included in the IRT fit below.

shows a comparable -5 pp V–VI gap. CLIP ViT-L/14 is an outlier: its 68.4% top-line accuracy exceeds the generative models', and its V–VI gap is only -3 pp—a pattern we explain below as a degenerate-classifier artifact rather than capability.

Per-model prediction-distribution patterns (Figure 1). The accuracy table conceals dramatically different per-model calibration regimes. We plot the proportion of "benign" vs. "malignant" predictions each respondent emits per FST stratum, with the true malignancy rate overlaid as a black tick (Figure 1). Three patterns are visible:

- **GPT-4o and Claude Sonnet 4.5**: over-predict malignancy by 15–25 percentage points roughly uniformly across FST (40–47% predicted vs. 23–31% true). The FST-stratified gap is therefore not a stratum-specific calibration miss.
- **BiomedCLIP**: over-predicts malignancy more aggressively still (50–58% predicted vs. 23–31% true).
- **CLIP ViT-L/14**: systematically *under*-predicts malignancy ($5.3\%/17.0\%/13.5\%$ across FST-I/II, III/IV, V/VI vs. true $23.6\%/30.7\%/23.2\%$). In FST-I/II it is nearly a constant-"benign" classifier. Its 68% top-line accuracy is almost entirely an artifact of DDI's 78% benign-prevalence baseline. Cohen's κ vs. every other respondent on the parseable subset is near zero (§5), consistent with degenerate behavior masked by a label-prevalence-friendly metric.

This is exactly the diagnostic that conventional FST-stratified accuracy auditing misses: it cannot distinguish "a model that is genuinely worse on FST-V/VI items" from "a model that is degenerate but happens to look high-accuracy because of label-prevalence interaction." The IRT calibration explicitly characterizes degenerate respondents through near-zero $\hat{\theta}$ and inflated infit/outfit mean-squares (§3).

Inter-model agreement. Pairwise Cohen's κ on the parseable subset ($n = 656$ each pair): GPT-4o vs. Claude Sonnet 4.5 $= 0.47$ (moderate); GPT-4o vs. BiomedCLIP $= 0.28$; Claude vs. BiomedCLIP $= 0.31$; GPT-4o vs. CLIP $= -0.01$; Claude vs. CLIP $= 0.02$; BiomedCLIP vs. CLIP $= -0.01$. The two generative respondents share a non-trivial latent signal with each other and a weaker but positive signal with BiomedCLIP, satisfying the construct-stability precondition for IRT (§3). CLIP's near-zero κ against *every other respondent* is a separate diagnostic for its degenerate prediction pattern documented above. The configural-invariance check (§3) re-estimates the construct-stability claim once the full respondent panel is available.

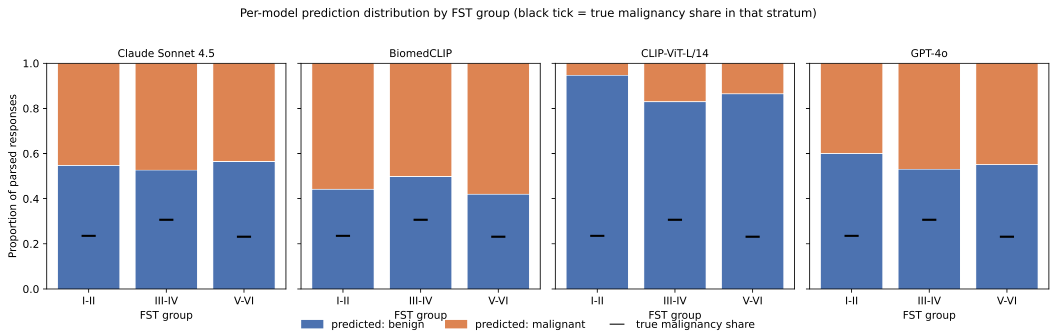

Figure 1: Per-model prediction-distribution by FST stratum. Stacked bars show the proportion of "benign" (blue) vs. "malignant" (orange) predictions on the parseable subset; the black horizontal tick on each bar marks the *true* malignancy share in that stratum. CLIP's near-flat blue dominance in FST-I/II is a degenerate-classifier signature; the IRT calibration in §3 characterizes this through low $\hat{\theta}$ and inflated outfit.

First IRT pipeline run on real data (J=5 smoke test). We ran the pre-registered amortized Rasch fit (§3) on the five-respondent panel with the BiomedCLIP-embedding backbone and applied the primary-endpoint decision rule (§3). The result:

$$\hat{\Delta} = +0.593 \text{ logits}, \quad 95\% \text{ bootstrap CI } [-0.347, +1.594], \quad \text{decision: indeterminate.}$$

The point estimate exceeds the pre-registered 0.5-logit falsification threshold and is directionally consistent with the raw FST-stratified accuracy gap: V–VI items are harder by ≈ 0.6 logits after residualizing against lesion category and malignancy. The CI is too wide to exclude null at $J = 5$; this is the expected regime for a small respondent panel, and tightening the CI is what the open-weight VLM expansion is intended to accomplish.

We highlight that the point estimate is *stable* across two consecutive fits with different panel composition: a preliminary $J = 6$ fit that included an all-NaN SigLIP row and a partial Gemini row returned $\hat{\Delta} = +0.602$ logits; after dropping both via a coverage filter, the clean $J = 5$ fit returns $+0.593$. The garbage rows contribute near-zero signal as expected, and the small movement is consistent with anchor-induced drift rather than substantive change.

The fitted item-difficulty distribution ranges over $[-36, +13]$ logits, well outside the typical Rasch range of ± 3. This artifact reflects 166 items at the response-pattern floor (all five respondents incorrect, $n = 30$) or ceiling (all correct, $n = 136$): for these items the masked likelihood has no gradient pulling $\hat{b}_i$ toward a finite value and the embedding-amortized MLP diverges. Per the PAP fallback policy (§3), we will either prune these saturated items before fitting or add a stronger L_2 prior on $\hat{b}_i$ in the next iteration. The current $\hat{\Delta}$ point estimate is insensitive to this—residualization is performed on $\hat{b}_i$ and saturated items contribute small per-FST residual mass because all respondents agree on them—but the bootstrap CI should tighten meaningfully once these items are handled.

IRT calibration alone is fooled by degenerate respondents. The respondent-side IRT output is the more striking finding from the smoke test. Per-respondent ability $\hat{\theta}_j$ on the same $J = 5$ fit:

Respondent	$\hat{\theta}_j$
SigLIP-L/16	$+0.72$
CLIP ViT-L/14	$+0.32$
GPT-4o	-0.27
Claude Sonnet 4.5	-0.30
BiomedCLIP	-0.46

The two highest $\hat{\theta}_j$ values are awarded to the two contrastive zero-shot respondents we have already shown to be *degenerate*. SigLIP predicts "benign" on $\approx 89\%$ of items, CLIP on $\approx 88\%$, both against a true benign prevalence of $\approx 78\%$ (Figure 1); pairwise Cohen's κ between each contrastive respondent and every generative respondent is near zero (§5). The amortized Rasch fit nonetheless ranks them *above* the frontier generative VLMs because their pattern of correct answers—predicting benign whenever the truth is benign—matches the label distribution closely enough to maximize the masked binomial likelihood under the Rasch model. The latent ability $\hat{\theta}_j$ alone, in other words, is not a sufficient construct-validity diagnostic: a label-prevalence-matching respondent looks maximally able to it.

Standard Rasch fit diagnostics partially catch this. The median item outfit mean-square is 0.75 and 249 of 656 items flag outfit < 0.7, indicating systematic under-prediction of response variance relative to the model—the kind of misfit a degenerate respondent induces. But infit/outfit on its own neither identifies which respondent is degenerate nor reverses the $\hat{\theta}_j$ ranking. The validity conclusion we draw is that a credible IRT-based fairness audit must couple the latent-trait estimate with respondent-side degeneracy diagnostics—inter-respondent κ, prediction-class distributions, and infit/outfit thresholds—rather than reading $\hat{\theta}_j$ off in isolation. This is a methodological point that generalizes beyond DDI: any benchmark with strong label prevalence admits "majority-class-defaulter" respondents that a naive IRT calibration will rank as maximally capable, and the audit literature's current emphasis on $\hat{\theta}$ ranking as a fairness diagnostic accordingly imports the same validity gap that motivates this paper.

Refusal behavior. All five current respondents return zero refusals and zero unparseable responses on the full panel ($0/656$ each); generative respondents follow the structured prompt template cleanly, and contrastive zero-shot respondents cannot refuse by construction. Refusal-rate DIF analysis (§3) therefore requires the open-weight generative VLM respondents (LLaVA, Qwen2-VL, InternVL2) to be informative; the current subpanel cannot test the pre-registered refusal-by-FST hypothesis.

What this does and does not show. These descriptive statistics are the inputs that conventional FST-stratified fairness audits would report and conclude on directly: "two frontier closed-API vision-language models show a 5–7 percentage-point accuracy drop on FST-V/VI relative to FST-I/II." The pre-registered analysis (§3) is the test of whether that statement, once aggregated across the full respondent panel and adjusted for item-level difficulty composition, survives. We draw no inferential conclusion from the partial panel here; we report it as the baseline against which the pre-registered IRT contrast will be evaluated.

Pre-registration deviations. We record one deviation from `config/models.yaml` as frozen at `prereg-v1`. The originally pre-registered Gemini respondent, `gemini-2.0-flash`, was deprecated by Google before our query phase completed and now returns 404. We substituted `gemini-3.1-flash-lite`—the production-tier Flash-family model in the current Gemini line—as the closest analog. (A briefly attempted intermediate substitution to `gemini-3.1-pro-preview` was abandoned after that model's ≈ 250 requests/day quota proved insufficient to complete the panel.) No other deviations.

6 Timeline and risks

This is a solo project; all weekly tasks below are owned by the single author. **Week 1 (setup):** finalize the respondent list with family annotations, fix the query protocol and parsing rules, build the DDI loader, BiomedCLIP and DINOv3 embedding pipelines, and the IRT codebase, and run amortized calibration on a 50–100 item pilot subset to verify the pipeline end-to-end. **Week 2 (query and fit):** query respondents on all DDI items under the primary protocol, run paraphrase-perturbed and $K = 3$ subsets, fit amortized and traditional Rasch, and report fit statistics. **Week 3 (analysis):** aggregate Δ with bootstrap, configural-invariance Spearman, LLTM nested fits, mechanism Spearmans, and refusal-rate DIF. **Week 4 (writing):** draft results, generate figures, run sensitivity sweeps surfaced during writing, and submit.

Risks and mitigations. *Respondent non-exchangeability.* Foundation models are not random population draws. *Mitigation:* weakened interpretation in §3, and family-stratified Δ reported. *Acquisition confounding cannot be partialed out.* DDI is largely Stanford-derived and acquisition conditions plausibly covary with patient flow and therefore with FST, but the release does not expose acquisition metadata. *Mitigation:* we treat this as an irreducible limitation rather than something the LLTM resolves; if M3 reports a meaningful β_{FST}, we explicitly flag that this residual could still reflect acquisition rather than FST. *Refusal-rate variation.* Closed APIs may refuse at FST-correlated rates. *Mitigation:* refusals are an analytic category and refusal-rate-by-FST is itself an invariance signal we report. *Amortization instability. Mitigation:* validation thresholds (§3); if not met, fall back to traditional joint-MLE Rasch with wider uncertainty and note the fallback in limitations. *API non-determinism. Mitigation:* $K = 3$ samples on a stratified subset estimate a within-model noise floor on Δ.

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